

Course project: Learning dynamical systems from data

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Project statement

Given the trajectory data of a single pendulum, we would like to predict its future states using FNNs, SympNets [1] and PINNs [2]. The governing equation is shown below:

$$\begin{aligned}\frac{dp_\theta}{dt} &= -mgl \sin \theta \\ \frac{d\theta}{dt} &= \frac{p_\theta}{ml^2}\end{aligned}\tag{1}$$

where g is the magnitude of the gravitational field, l is the length of the rod or cord, and θ is the angle from the vertical to the pendulum, p_θ is the angular momentum of the pendulum.

Dataset

Consider a nondimensionalized system with $m = l = g = 1$. Then set the initial conditions $[p_\theta(0), \theta(0)] = [0, 1]$. The data for θ and p_θ on $n = 140$ time points are generated by a Stormer–Verlet integrator and provided in the plain text file (train.txt and test.txt), denoted by $\{x_i\}_1^{140}$. The data are sampled on a uniform grid of stepsize $h = 0.1$. The first column in the txt file represents p_θ and the second column represents θ .

Important: You may use LLMs for *programming assistance only*. All mathematical derivations, diagnostics, interpretations, and explanations must be your own. Your grade will be based primarily on conceptual understanding and the quality of your reasoning.

Task 0: Physics sanity checks (required)

Before training any model, do the following.

1. Write the *Hamiltonian* $H(\theta, p_\theta)$ for (??). Show that the system can be written in canonical Hamiltonian form with the standard symplectic matrix.
2. Identify at least two invariants/qualitative properties of the true dynamics that you expect a good long-time predictor to respect (e.g., energy behavior, periodicity, phase-space geometry).
3. Using the provided data, compute the discrete-time energy $H(\theta_i, p_{\theta,i})$ and plot it vs. time. Briefly interpret what you see (numerical drift, oscillations, etc.).

Task 1: FNN — When a good one-step model fails long-term

1. Train a surrogate FNN model on the dataset to learn the discrete flow map $x_{i-1} \mapsto x_i$, where $x = (p_\theta, \theta)$. Use 40 points for training and the remaining 100 points for testing. For the training set, prepare input-target pairs $\mathcal{T} = \{(x_{i-1}, x_i)\}_{i=1}^{40}$.
2. Perform *two* evaluations:
 - **Teacher-forcing error:** evaluate the one-step prediction error on the 100 test pairs.
 - **Rollout error:** recursively apply the model to generate $\tilde{x}_{41}, \dots, \tilde{x}_{140}$.

Plot the phase portrait of $\tilde{x}$ vs. x and the MSE vs. time.

3. Provide a short explanation (with evidence) answering: *How can a model have low one-step error but poor rollout behavior?* Your explanation must explicitly discuss error accumulation and the geometry of the phase portrait.
4. Diagnose whether the FNN preserves (approximately) any invariant from Task 0. Support your claim with at least one quantitative diagnostic beyond MSE.

Task 2: SympNet — Structure as an inductive bias

1. Replace the FNN in Task 1 with a SympNet and repeat the training and evaluation.
2. Explain precisely *which* structure SympNet enforces in the learned map. State what “symplectic” means for a discrete map in 2D phase space, and propose a numerical test for symplecticity of your learned map (e.g., Jacobian-based test or area-preservation test). Apply your test.
3. Compare energy behavior in rollout between FNN and SympNet (use the Hamiltonian you derived). Discuss whether better energy behavior necessarily implies better phase accuracy, or vice versa.
4. Make a specific, falsifiable claim about why SympNet should (or should not) improve long-time prediction for this dataset, and verify it with an experiment.

Task 3: PINN — Parameter estimation and identifiability

1. Formulate a PINN for estimating m and l from the same training data, keeping $g = 1$ fixed.
2. **Identifiability question (core):** Argue whether m and l are individually identifiable from trajectory data of $\theta(t)$ and $p_\theta(t)$ when g is fixed, or whether only certain combinations (e.g., ml^2 or mgl) are identifiable. Your argument must start from (??) and should include a clear conclusion.
3. Train the PINN, report the learned parameters, and then integrate (??) forward in time with a Störmer–Verlet integrator using the learned parameters. Compare the predicted and true trajectories in phase space and compare MSE vs. time.
4. Perform a *sensitivity* experiment: perturb the learned parameters by $\pm 1\%$ (one at a time, holding the other fixed) and show how the long-time trajectory changes. Use this to discuss “parameter error” vs “trajectory error”.

Task 4: Comparative analysis (conceptual)

Provide a concise but rigorous discussion addressing:

- Rank FNN, SympNet, and PINN in terms of *modeling bias*, *inductive bias*, *data efficiency*, and *robustness to distribution shift* (e.g., different initial condition).
- Does imposing stronger physical structure always improve long-time prediction? Use your experiments to support your answer.
- Suppose the data contain additive noise. Which approach would you trust most and why? Propose a concrete modification to improve robustness for your chosen method.

Submission Requirements

- A short report (6–8 pages) emphasizing diagnostics, reasoning, and interpretation.
- Include: (i) one-step vs rollout error comparison, (ii) phase portraits, (iii) energy behavior, and (iv) your identifiability argument.
- Code may be assisted by LLMs; the written analysis must be your own.

Programming Options

You may use TensorFlow, PyTorch, DeepXDE or Modulus for completing the tasks.

References

- [1] P. Jin, Z. Zhang, A. Zhu, Y. Tang, and G. E. Karniadakis. Sympnets: Intrinsic structure-preserving symplectic networks for identifying hamiltonian systems. *Neural Networks*, 132:166–179, 2020.
- [2] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.